

Quiz Cheating Detection – Graduation Project Overview

My graduation project focuses on detecting cheating during online exams using the following detectors. Any detection triggers a red flag:

1. **Tab Switching:** If the student opens a new tab or navigates away from the exam tab.
2. **Object Detection via Camera:** If the student's camera detects prohibited objects (e.g., mobile phone, book) during the exam.
3. **Eye Gaze Monitoring:** If the student's eyes look away from the screen for a defined duration (e.g., 15 seconds).
4. **Multi-Person Detection:** If the camera detects more than one person in the frame.

Each of these events is logged as a cheating indicator.

System Workflow

The system supports two user roles: **Instructor** and **Student**.

Instructor Role

- **Exam Creation:**
The instructor can create exams with the following question types:
 - **MCQ (Multiple Choice Questions):** Single-answer or multi-answer.
 - **Q/A (Open-ended questions).**
 - **Exams may include a mix of both MCQ and Q/A questions.**
- **Exam Access:**
After creating the exam, a unique **exam code** is generated. Students use this code to join the exam.
- **Results Review:**
After the exam ends, the instructor can access a **Results page** that displays:
 - Each student's score.
 - MCQ answers are **automatically graded**.
 - Q/A answers require **manual grading** by the instructor.
 - A **status** for each student based on detected cheating behavior:
 - **Clean:** No suspicious activity detected.
 - **Warning:** One cheating detector was triggered.
 - **Red Flag:** More than one detector was triggered.
- **Student Report (Details):**
Next to each student, there is a **"Details"** button. Clicking it opens a detailed report showing:
 - Time spent on each question.
 - Any cheating detectors triggered during each question.

- Based on this information, the instructor can **re-evaluate or adjust** the student's grade if needed.
 - **Finalization & Notification:**
Once all students are graded, the instructor finalizes the results, and the system **sends the results to the students.**
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Student Role

- **Account & Access:**
The student creates an account with the **Student** role. They **cannot create exams—only take them.**
- **Taking the Exam:**
 - The student enters an exam using the provided **exam code.**
 - Before the exam starts, they see **a warning page** listing exam rules (e.g., “Do not cheat”).
 - The student **must grant camera permission**, but they are **not explicitly told** that their behavior is being monitored for cheating. The system only requests camera access and displays general rules.
- **Viewing Results:**
After the instructor finalizes grades, the student can view results for all **previous exams** they have taken.